

Budget Management System

A Python & MySQL Based Personal Budget Management System

1. Introduction

Budget Management System is a Python and MySQL based application developed to help users manage their personal finances efficiently. The system allows users to record income and expenses, set monthly budgets, monitor savings, generate reports, and receive budget alerts. All data is stored permanently in a MySQL database, making the system reliable and easy to use.

2. Objectives

- To maintain personal income and expense records.
 - To manage monthly budgets efficiently.
 - To calculate savings automatically.
 - To generate financial reports.
 - To provide alerts when the budget limit is exceeded.
 - To develop a database-driven application using Python and MySQL.
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3. Technologies Used

Component	Technology
Programming Language	Python 3
Database	MySQL
Database Connector	mysql-connector-python
Interface	Console Application
IDE	Visual Studio Code

4. Modules of the System

Module 1: User Management

Functions:

- User Registration
- User Login
- Change Password
- User Profile

Table Used:

- users
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Module 2: Category Management

Functions:

- Add Category
- View Categories
- Update Category
- Delete Category

Examples:

- Salary
- Food
- Rent
- Shopping
- Travel

Table Used:

- category
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Module 3: Income Management

Functions:

- Add Income
- View Income
- Update Income

- Delete Income

Table Used:

- income
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Module 4: Expense Management

Functions:

- Add Expense
- View Expense
- Update Expense
- Delete Expense

Table Used:

- expense
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Module 5: Budget Management

Functions:

- Set Monthly Budget
- Update Budget
- View Budget
- Remaining Budget

Table Used:

- budget
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Module 6: Savings Management

Functions:

- Calculate Savings
- Monthly Savings
- Total Savings

Formula:

Savings = Total Income – Total Expense

Table Used:

- savings
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Module 7: Reports

Functions:

- Monthly Income Report
 - Monthly Expense Report
 - Category-wise Expense Report
 - Total Income
 - Total Expense
 - Total Savings
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Module 8: Alerts & Notifications

Functions:

- Budget Exceeded Alert
 - Low Balance Alert
 - Monthly Reminder
 - Savings Goal Notification
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5. Database Design

The project contains the following tables:

1. users
2. category
3. income
4. expense
5. budget
6. savings

Relationships:

- One user can have multiple income records.
- One user can have multiple expense records.
- One category can be used for many income or expense records.
- Each user can set a monthly budget.
- Savings are calculated from income and expense records.

6. System Workflow

1. User Login/Register
2. Select Category
3. Add Income
4. Add Expense
5. Set Monthly Budget
6. Calculate Savings
7. View Reports
8. Receive Budget Alerts

7. Future Enhancements

- Graphical User Interface (GUI)
- Mobile Application
- Cloud Database
- PDF Report Generation
- Email Notifications
- Data Backup and Restore

8. Conclusion

The Budget Management System is a simple and efficient application developed using Python and MySQL. It helps users manage their income, expenses, budgets, and savings in an organized manner. The system provides useful reports and alerts, making personal financial management easier. The project demonstrates the practical use of Python programming, MySQL database connectivity, CRUD operations, and database management concepts.